Qp code. 25901546

MAHATMA GANDHI UNIVERSITY , KOTTAYAM

MGU- UGP (HONOURS) Regular examination March 2026

FOURTH SEMESTER

(2024 ADMISSION ONWARDS)

STATISTICS

MG4DSCSTA201 - Statistical Inference

SCHEME OF EVALUATION **Maximum marks: 50**

Part A

Short Answer Questions

Answer any seven questions. Each question carries two marks

1. Chebychev's inequality for finding upper bound of probability- 2 marks
2. Efficiency Comparison - 1 mark
   Finding relative efficiency of t_1 with respect to t_2 - 1 mark
3. Define Unbiasedness- 2 marks
4. Finding Minimum variance unbiased estimator of σ^2 - 1 mark
   Finding value of minimum variance—1 mark
   (or definition of Minimum variance unbiased estimator only - 1 mark)
5. Finding maximum likelihood estimator for parameter θ – 2 marks
   (or definition of maximum likelihood estimator only – 1 mark)
6. Explanation of likelihood ratio in Neyman Pearson most powerful test procedure- 2 marks
   (or Neyman Pearson most powerful test -statement- 2 marks)
7. Expected Frequency= (Row total x Column total)/ grand total -- 2 marks
8. Explanation—2 marks
9. Correct test statistic- 2 marks
10. Correct test statistic- 2 marks (7*2=14)

PART B

Short Essay Questions

Answer any four questions. Each question carries six marks.

11. Proof -6 marks
    (or weak law of large numbers (statement only) – 2 marks)
12. Derivation of confidence interval – 6 marks
    (or confidence interval only – 2 marks)

13. Correct Explanation- 6 marks

14. Correct Explanation- 6 marks

15. Define Level of significance- 1 mark

    Define Power of test -1 mark

    Level of significance=0.05- 2 marks

    Power of test =0.31 – 2 marks

16. H_0: The data belongs to normal distribution. - 0.5 mark
    H_1: The data doesn't belong to normal distribution. - 0.5 mark
    Anderson Darling test statistic – 2 marks, Anderson Darling test statistic value= 0.175842- 2 marks
    Conclusion- Accept H_0. – 1 mark

(4*6=24)

PART C
Essay Questions
Answer any one question. Each question carries 12 marks.

17. Explanation of maximum likelihood method of estimation – 5 marks

    Properties of maximum likelihood estimators - 2 marks

    Explanation of moment method of estimation - 4 marks

    Advantage of moment method of estimation- 1 mark

18. Explanation of test procedure of one way ANOVA- 12 marks

(1*12=12)